

```
In [11]: import pandas as pd
df = pd.read_csv(r"C:\Users\jahan\OneDrive\Documents\customer_shopping_behavior.csv")
```

```
In [36]: df.head(10)
```

```
Out[36]:
```

	customer_id	age	gender	item_purchased	category	purchase_amount	location
0	1	55	Male	Blouse	Clothing	53	Kentucky
1	2	19	Male	Sweater	Clothing	64	Maine
2	3	50	Male	Jeans	Clothing	73	Massachusetts
3	4	21	Male	Sandals	Footwear	90	Rhode Island
4	5	45	Male	Blouse	Clothing	49	Oregon
5	6	46	Male	Sneakers	Footwear	20	Wyoming
6	7	63	Male	Shirt	Clothing	85	Montana
7	8	27	Male	Shorts	Clothing	34	Louisiana
8	9	26	Male	Coat	Outerwear	97	West Virginia
9	10	57	Male	Handbag	Accessories	31	Missouri



```
In [13]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
In [17]: df.describe()
```

Out[17]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

```
In [18]: df.isnull().sum()
```

```
Out[18]: Customer ID      0
         Age              0
         Gender           0
         Item Purchased   0
         Category         0
         Purchase Amount (USD) 0
         Location         0
         Size             0
         Color            0
         Season           0
         Review Rating    37
         Subscription Status 0
         Shipping Type    0
         Discount Applied 0
         Promo Code Used  0
         Previous Purchases 0
         Payment Method   0
         Frequency of Purchases 0
         dtype: int64
```

```
In [19]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x:x.
```

```
In [20]: df.isnull().sum()
```

```
Out[20]: Customer ID      0
         Age              0
         Gender           0
         Item Purchased   0
         Category         0
         Purchase Amount (USD) 0
         Location         0
         Size             0
         Color            0
         Season           0
         Review Rating    0
         Subscription Status 0
         Shipping Type    0
         Discount Applied 0
         Promo Code Used  0
         Previous Purchases 0
         Payment Method   0
         Frequency of Purchases 0
         dtype: int64
```

```
In [24]: df.columns = df.columns.str.lower()
         df.columns = df.columns.str.replace(' ', '_')
         df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [25]: df.columns
```

```
Out[25]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
              'purchase_amount', 'location', 'size', 'color', 'season',  
              'review_rating', 'subscription_status', 'shipping_type',  
              'discount_applied', 'promo_code_used', 'previous_purchases',  
              'payment_method', 'frequency_of_purchases'],  
             dtype='object')
```

```
In [30]: #create a column age_group  
labels = ['young_adults', 'adults', 'middle age', 'senior']  
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
In [34]: df[['gender', 'age', 'age_group']].head(10)
```

```
Out[34]:
```

	gender	age	age_group
0	Male	55	middle age
1	Male	19	young_adults
2	Male	50	middle age
3	Male	21	young_adults
4	Male	45	middle age
5	Male	46	middle age
6	Male	63	senior
7	Male	27	young_adults
8	Male	26	young_adults
9	Male	57	middle age

```
In [54]: frequency_mapping = {  
        'Fortnightly': 14,  
        'Weekly': 7,  
        'Monthly': 30,  
        'Quarterly': 90,  
        'Bi-Weekly': 14,  
        'Annually': 365,  
        'Every 3 Months': 90  
        }  
  
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
In [57]: df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

```
Out[57]:
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
In [63]: df.columns
```

```
Out[63]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
               'purchase_amount', 'location', 'size', 'color', 'season',  
               'review_rating', 'subscription_status', 'shipping_type',  
               'discount_applied', 'previous_purchases', 'payment_method',  
               'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
              dtype='object')
```

```
In [83]: %load_ext sql
```

ModuleNotFoundError

Traceback (most recent call last)

Cell In[83], line 1

```
----> 1 get_ipython().run_line_magic('load_ext', 'sql')
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\IPython\core\interactiveshell.py:2480, in InteractiveShell.run_line_magic(self, magic_name, line, _stack_depth)

```
    2478     kwargs['local_ns'] = self.get_local_scope(stack_depth)
    2479     with self.builtin_trap:
-> 2480         result = fn(*args, **kwargs)
    2482     # The code below prevents the output from being displayed
    2483     # when using magics with decorator @output_can_be_silenced
    2484     # when the last Python token in the expression is a ';'.
    2485     if getattr(fn, magic.MAGIC_OUTPUT_CAN_BE_SILENCED, False):
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\IPython\core\magics\extension.py:33, in ExtensionMagics.load_ext(self, module_str)

```
    31     if not module_str:
    32         raise UsageError('Missing module name.')
---> 33     res = self.shell.extension_manager.load_extension(module_str)
    35     if res == 'already loaded':
    36         print("The %s extension is already loaded. To reload it, use:" % module_
str)
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\IPython\core\extensions.py:62, in ExtensionManager.load_extension(self, module_str)

```
    55     """Load an IPython extension by its module name.
    56
    57     Returns the string "already loaded" if the extension is already loaded,
    58     "no load function" if the module doesn't have a load_ipython_extension
    59     function, or None if it succeeded.
    60     """
    61     try:
---> 62         return self._load_extension(module_str)
    63     except ModuleNotFoundError:
    64         if module_str in BUILTINS_EXTS:
```

File ~\AppData\Local\Programs\Python\Python312\Lib\site-packages\IPython\core\extensions.py:77, in ExtensionManager._load_extension(self, module_str)

```
    75     with self.shell.builtin_trap:
    76         if module_str not in sys.modules:
---> 77             mod = import_module(module_str)
    78             mod = sys.modules[module_str]
    79             if self._call_load_ipython_extension(mod):
```

File ~\AppData\Local\Programs\Python\Python312\Lib\importlib__init__.py:90, in import_module(name, package)

```
    88         break
    89         level += 1
---> 90     return _bootstrap._gcd_import(name[level:], package, level)
```

File <frozen importlib._bootstrap>:1387, in _gcd_import(name, package, level)

File <frozen importlib._bootstrap>:1360, in _find_and_load(name, import_)

```
File <frozen importlib._bootstrap>:1324, in _find_and_load_unlocked(name, import_)
```

```
ModuleNotFoundError: No module named 'sql'
```

```
In [84]: %sql mysql+pymysql://root:Ritesh@21092005@localhost
```

```
UsageError: Line magic function `%sql` not found.
```

```
In [86]: %%sql
```

```
SELECT VERSION();
```

```
UsageError: Cell magic `%%sql` not found.
```

```
In [ ]:
```

```
In [88]: import pandas as pd
        from sqlalchemy import create_engine
```

```
In [89]: engine = create_engine(
        "mysql+pymysql://root:Ritesh%4021092005@localhost/customer_db"
        )
```

```
In [91]: pip install ipython-sql
```

Collecting ipython-sql
 Using cached ipython_sql-0.5.0-py3-none-any.whl.metadata (17 kB)
 Collecting prettytable (from ipython-sql)
 Using cached prettytable-3.18.0-py3-none-any.whl.metadata (37 kB)
 Requirement already satisfied: ipython in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython-sql) (8.28.0)
 Requirement already satisfied: sqlalchemy>=2.0 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython-sql) (2.0.51)
 Collecting sqlparse (from ipython-sql)
 Using cached sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
 Requirement already satisfied: six in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython-sql) (1.16.0)
 Collecting ipython-genutils (from ipython-sql)
 Using cached ipython_genutils-0.2.0-py2.py3-none-any.whl.metadata (755 bytes)
 Requirement already satisfied: greenlet>=1 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from sqlalchemy>=2.0->ipython-sql) (3.5.3)
 Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from sqlalchemy>=2.0->ipython-sql) (4.13.2)
 Requirement already satisfied: decorator in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (5.1.1)
 Requirement already satisfied: jedi>=0.16 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (0.19.1)
 Requirement already satisfied: matplotlib-inline in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (0.1.7)
 Requirement already satisfied: prompt-toolkit<3.1.0,>=3.0.41 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (3.0.48)
 Requirement already satisfied: pygments>=2.4.0 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (2.18.0)
 Requirement already satisfied: stack-data in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (0.6.3)
 Requirement already satisfied: traitlets>=5.13.0 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (5.14.3)
 Requirement already satisfied: colorama in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from ipython->ipython-sql) (0.4.6)
 Requirement already satisfied: wcwidth in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from prompt-toolkit<3.1.0,>=3.0.41->ipython->ipython-sql) (0.2.13)
 Requirement already satisfied: parso<0.9.0,>=0.8.3 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from jedi>=0.16->ipython->ipython-sql) (0.8.4)
 Collecting wcwidth (from prompt-toolkit<3.1.0,>=3.0.41->ipython->ipython-sql)
 Using cached wcwidth-0.8.2-py3-none-any.whl.metadata (43 kB)
 Requirement already satisfied: executing>=1.2.0 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from stack-data->ipython->ipython-sql) (2.1.0)
 Requirement already satisfied: asttokens>=2.1.0 in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from stack-data->ipython->ipython-sql) (2.4.1)
 Requirement already satisfied: pure-eval in C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages (from stack-data->ipython->ipython-sql) (0.2.3)
 Using cached ipython_sql-0.5.0-py3-none-any.whl (20 kB)
 Using cached ipython_genutils-0.2.0-py2.py3-none-any.whl (26 kB)
 Using cached prettytable-3.18.0-py3-none-any.whl (37 kB)
 Using cached wcwidth-0.8.2-py3-none-any.whl (323 kB)

Using cached sqlparse-0.5.5-py3-none-any.whl (46 kB)

Installing collected packages: ipython-genutils, wcwidth, sqlparse, prettytable, ipython-sql

```
----- 0/5 [ipython-genutils]
Attempting uninstall: wcwidth
----- 0/5 [ipython-genutils]
Found existing installation: wcwidth 0.2.13
----- 0/5 [ipython-genutils]
Uninstalling wcwidth-0.2.13:
----- 0/5 [ipython-genutils]
Successfully uninstalled wcwidth-0.2.13
----- 0/5 [ipython-genutils]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 1/5 [wcwidth]
----- 2/5 [sqlparse]
----- 2/5 [sqlparse]
----- 2/5 [sqlparse]
----- 3/5 [prettytable]
----- 4/5 [ipython-sql]
----- 5/5 [ipython-sql]
```

Successfully installed ipython-genutils-0.2.0 ipython-sql-0.5.0 prettytable-3.18.0 sqlparse-0.5.5 wcwidth-0.8.2

Note: you may need to restart the kernel to use updated packages.

In [92]: `%load_ext sql`

In [93]: `%sql mysql+pymysql://root:Ritesh%4021092005@localhost/customer_db`

```
File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sqlalchemy\engine\default.py", line 630, in connect
    return self.loaded_dbapi.connect(*cargs, **cparams) # type: ignore[no-any-return] # NOQA: E501
```



```
alchemy\engine\base.py", line 3319, in raw_connection
    return self.pool.connect()
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 448, in connect
    return _ConnectionFairy._checkout(self)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 1272, in _checkout
    fairy = _ConnectionRecord.checkout(pool)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 712, in checkout
    rec = pool._do_get()
          ^^^^^^^^^^^^^^
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\impl.py", line 178, in _do_get
    with util.safe_reraise():
         File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\util\langhelpers.py", line 122, in __exit__
         raise exc_value.with_traceback(exc_tb)
         File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\impl.py", line 176, in _do_get
         return self._create_connection()
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 389, in _create_connection
    return _ConnectionRecord(self)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 674, in __init__
    self.__connect()
         File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 900, in __connect
         with util.safe_reraise():
             File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\util\langhelpers.py", line 122, in __exit__
             raise exc_value.with_traceback(exc_tb)
             File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\pool\base.py", line 896, in __connect
                 self.dbapi_connection = connection = pool._invoke_creator(self)
                                                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\engine\create.py", line 667, in connect
    return dialect.connect(*cargs_tup, **cparams)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
alchemy\engine\default.py", line 630, in connect
    return self.loaded_dbapi.connect(*cargs, **cparams) # type: ignore[no-any-retur
n] # NOQA: E501
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\connections.py", line 372, in __init__
    self.connect()
         File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\connections.py", line 702, in connect
```

```

        self._request_authentication()
    File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\connections.py", line 1022, in _request_authentication
        auth_packet = _auth.caching_sha2_password_auth(self, auth_packet)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\_auth.py", line 245, in caching_sha2_password_auth
        pkt = conn._read_packet()
        ^^^^^^^^^^^^^^^^^^^^^^^^^
    File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\connections.py", line 803, in _read_packet
        packet.raise_for_error()
    File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\protocol.py", line 219, in raise_for_error
        err.raise_mysql_exception(self._data)
    File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\pym
ysql\err.py", line 154, in raise_mysql_exception
        raise errorclass(errno, errval, sqlstate=sqlstate)
sqlalchemy.exc.OperationalError: (pymysql.err.OperationalError) (1049, "Unknown data
base 'customer_db'")
(Background on this error at: https://sqlalche.me/e/20/e3q8)

```

Connection info needed in SQLAlchemy format, example:
 postgresql://username:password@hostname/dbname
 or an existing connection: dict_keys([])

```
In [94]: %%sql
        SELECT * FROM customers;
```

```

Traceback (most recent call last):
  File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
\magic.py", line 196, in execute
    conn = sql.connection.Connection.set(
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\jahan\AppData\Local\Programs\Python\Python312\Lib\site-packages\sql
\connection.py", line 82, in set
    raise ConnectionError(
sql.connection.ConnectionError: Environment variable $DATABASE_URL not set, and no c
onnect string given.

```

Connection info needed in SQLAlchemy format, example:
 postgresql://username:password@hostname/dbname
 or an existing connection: dict_keys([])

```
In [95]: %sql mysql+pymysql://root:Ritesh%4021092005@localhost/customer_db
```

```
In [96]: %%sql
        SELECT * FROM customers;

* mysql+pymysql://root:***@localhost/customer_db
(pymysql.err.ProgrammingError) (1146, "Table 'customer_db.customers' doesn't exist")
[SQL: SELECT * FROM customers;]
(Background on this error at: https://sqlalche.me/e/20/f405)

```

```
In [99]: import pandas as pd
        from sqlalchemy import create_engine
```

```

engine = create_engine(
    "mysql+pymysql://root:Ritesh%4021092005@localhost/customer_db"
)

query = "SHOW TABLES"

df = pd.read_sql(query, engine)

print(df)

```

Empty DataFrame
Columns: [Tables_in_customer_db]
Index: []

In [100...

```

import pandas as pd

df = pd.read_csv(r"C:\Users\jahan\OneDrive\Documents\customer_shopping_behavior.csv")

df.to_sql(
    r"C:\Users\jahan\OneDrive\Documents\customer_shopping_behavior.csv",
    engine,
    if_exists="replace",
    index=False
)

```

C:\Users\jahan\AppData\Local\Temp\ipykernel_21904\2500996576.py:5: UserWarning: The provided table name 'C:\Users\jahan\OneDrive\Documents\customer_shopping_behavior.csv' is not found exactly as such in the database after writing the table, possibly due to case sensitivity issues. Consider using lower case table names.

```
df.to_sql(
```

Out[100...] 3900

In [101...

```

import pandas as pd
from sqlalchemy import create_engine

engine = create_engine(
    "mysql+pymysql://root:Ritesh%4021092005@localhost/customer_db"
)

# Read CSV
df = pd.read_csv(r"C:\Users\jahan\OneDrive\Documents\customer_shopping_behavior.csv")

# Save into MySQL as a table
df.to_sql(
    "customer_shopping_behavior", # <-- Table name
    con=engine,
    if_exists="replace",
    index=False
)

print("Data imported successfully!")

```

Data imported successfully!

In []: